SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	*Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)*		
Student Name:	Y. Thanushma	Reg. No.:	192425355
Section / Batch:	2024/ Slot A/Hostel	Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT

1. A logistic company computes the distance between warehouses to identify the closest ones using Brute force computation

2. A payroll system uses Binary search on sorted employee ID's, but frequent insertions disturb order.

Code of Conduct

I Y. Thanushma /192425355 (Name / Reg No) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Y. Thanu

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature:	Signature:	Signature:
Date: 11/8/26	Date:	Date: 11/8/26

:- Y. Thanushma

er Number :- 192425355

urse Code :- CBA0617

Course Name :- Design and Analysis of Algorithms

Course Faculty :- Dr. Pradeep kumar

CO-2 Assessment 2

1) Closest Pair of Warehouse Using Brute Force

A logistic company needs to find the two closest warehouse to reduce transportation cost and delivery time. The Brute Force method checks every unique pair of warehouses and selects the pair with the minimum distance. It gaurantees the correct result for small datasets.

a) Explain how all pairs are evaluated

In the closest Brute Force algorithm, the logistic company calculates the distance between every possible pair of warehouses.

Steps :-

1. Take the first warehouse
2. Compare it with every other warehouse and calculate the distance.
3. Move the second warchouse and compare it with remaining warehouses.
4. Continue until all unique pairs have been checked
5. The pair with the minimum distance is identified as the closest pair.

For n warehouses, the total number of pairs evaluated is :-

$$\frac{n(n-1)}{2}$$

b) Derive the time complexity

For each warehouse, the algorithm compares it with the remaining warehouses.

Number of comparisons :-

$$(n-1) + (n-2) + \cdots + 2 + 1$$
$$= \frac{n(n-1)}{2}$$

Ignoring constants,

$$T(n) = O(n^2)$$

Compute the Euclidean distance :- $d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

Space Complexity :- $O(1)$

only a few variables are used.

c) Discuss Scalability Concerns :-

The brute force approached works well for a small number of warehouses, but it becomes inefficient as the number increases.

Scalability Issues :-

* Every warehouse is compared with every other warehouse.

...e number of distance calculations increases rapidly with large datasets.

* Execution time becomes very high for thousands or millions of warehouses.

* It is unsuitable for real-time logistics systems with frequent updates.

Example:-

100 warehouses → 4950 comparisons
1000 warehouses → 499500 comparisons
10,000 warehouses → 49995000 comparisons

Thus, the Brute Force algorithm does not scale well. For large datasets, more efficient algorithms such as the Divide and Conquer closest pair algorithm with $O(n \log n)$ time complexity are preferred.

$O(n^2)$ time complexity makes it inefficient for large-scale logistic applications.

2) Binary Search in a Payroll System

A payroll system stores employees ID's and uses Binary search to quickly find a specific find a Specific employee record. Binary search works efficiently only when the employee ID's are arranged in sorted order

a) Explain Binary search Operation

Binary search is used to search an employee ID in a sorted list. It repeatedly divides the search range into two halves until the required ID found.

Steps :-

* Find the emiddle employee ID
* If the middle ID matches the target, the search is complete.
* If the target ID is smaller, search the left half
* If the target ID is larger, search the right half
* Repeat the until the ID is found or the search range becomes empty.

For n employee ID's, the search requires approximately $\log_2 n$ comparisons.

b) Analyze the impact of Unsorted Data

Binary search works correctly only when employee ID's are sorted.

If frequent insertions disturb the sorted order:

* Binary search may return incorrect results.
* It may fail to find an existing employee ID
* The list must be sorted again or a linear search $(O(n))$ must be used

Time Complexity :-

Binary search (sorted data) : $O(\log n)$
Binary search (unsorted data) : Not applicable
Linear search (unsorted data) : $O(n)$

Discuss Maintaining Sorted Data for Efficiency

To keep Binary search efficient, the employee IDs should always remain sorted

Methods :-

* Insert each new employee ID in its correct sorted position
* Periodically sort the list after multiple insertions
* Use data structures such as Balanced Binary Search Trees (BSTs) or B-Trees, which automatically maintain sorted order.

Benifits :-

* Faster searching using Binary search
* Accurate search results.
* Better performance for payroll systems with frequent searches.

Binary Search is a fast and efficient searching algorithm with O(logn) time complexity, but it requires sorted employee IDs. Frequent insertions can disturb the order and reduce efficiency. Therefore, maintaining sorted data ensures accuracy and efficient payroll operations.